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const int trig=10;
const int echo=11;
int gled=3;
int yled=5;
int rled=6;
int pot=A3;
long duration;
int distance;

void setup()
{
  pinMode(gled, OUTPUT);
  pinMode(yled, OUTPUT);
  pinMode(rled, OUTPUT);
  pinMode(pot, INPUT);
  pinMode(trig, OUTPUT);
  pinMode(echo, INPUT);
  Serial.begin(9600);
}

void loop()
{
  digitalWrite(trig, LOW);
  delayMicroseconds(2);

  digitalWrite(trig, HIGH);
  delayMicroseconds(10);
  digitalWrite(trig, LOW);

  duration=pulseIn(echo,HIGH);
  distance=duration*0.034/2;

  Serial.print("Distance= ");
  Serial.print(distance);
  Serial.println(" cm");

  int input = analogRead(pot);
  int warningdistance=map(input,0,1023,10,50);

  Serial.print("Warning distance: ");
  Serial.print(warningdistance);
  Serial.println(" cm");
  if (distance>warningdistance)
  {
    digitalWrite(gled, HIGH);
    digitalWrite(rled, LOW);
    digitalWrite(yled, LOW);
    Serial.println("Robot is at a safe distance");
  }
  else if (distance>(warningdistance/2))
  {
    digitalWrite(yled, HIGH);
    digitalWrite(rled, LOW);
    digitalWrite(gled, LOW);
    Serial.println("Robot is getting close to the obstacle");
  }
  else
  {

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    digitalWrite(rled, HIGH);  
    digitalWrite(yled, LOW);  
    digitalWrite(gled, LOW);  
        Serial.println("Robot is dangerously close to the obstacle");  
    }  
    delay(500);  
}
```